

Mortality, Morbidity, and Population Health Measurement

Crude & standardized mortality · cause-of-death analysis · DALYs, HALE, and summary measures of health

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Learning Objectives

1

Compare mortality measures

Distinguish crude death rates from age-specific mortality rates and explain why the crude rate can mislead.

2

Standardize rates

Apply direct and indirect age standardization to compare mortality fairly across populations with different age structures.

3

Analyze cause of death

Interpret cause-of-death data, data-quality limits, and leading global causes of mortality and disease burden.

4

Apply summary health measures

Explain DALYs, YLL, YLD, and HALE, and use them to interpret global, Thai, and comparative Asian population health.

Agenda — 180-Minute Session Structure

- 1 Welcome & learning objectives
- 2 Block 1 — Crude vs. age-specific mortality; direct & indirect standardization (two worked examples each)
- 3 **BREAK**
- 4 Block 2 — Cause-of-death analysis, data quality, and the COVID-19 mortality shock, in retrospect
- 5 Block 3 — Summary measures of population health: DALYs, YLL/YLD, HALE (two worked examples)
- 6 In-class discussion & practice exercises
- 7 Summary & references

01

BLOCK 1 • 12:00–1:00

Measuring Mortality: Crude, Specific, and Standardized Rates

Why the simplest mortality measure is also the most misleading — and the two standardization methods demographers use to fix it.

THE STARTING POINT

The Crude Death Rate (CDR)

- Expressed as deaths per 1,000 population per year — the simplest mortality indicator, requiring only two numbers.
- The problem: the CDR mixes the population's actual mortality risk with its age structure. An older population will show a higher CDR than a younger one even if age-specific risks are identical or lower.
- It is still the most widely reported mortality statistic in the news media — precisely because it is easy to compute, not because it is easy to interpret correctly.

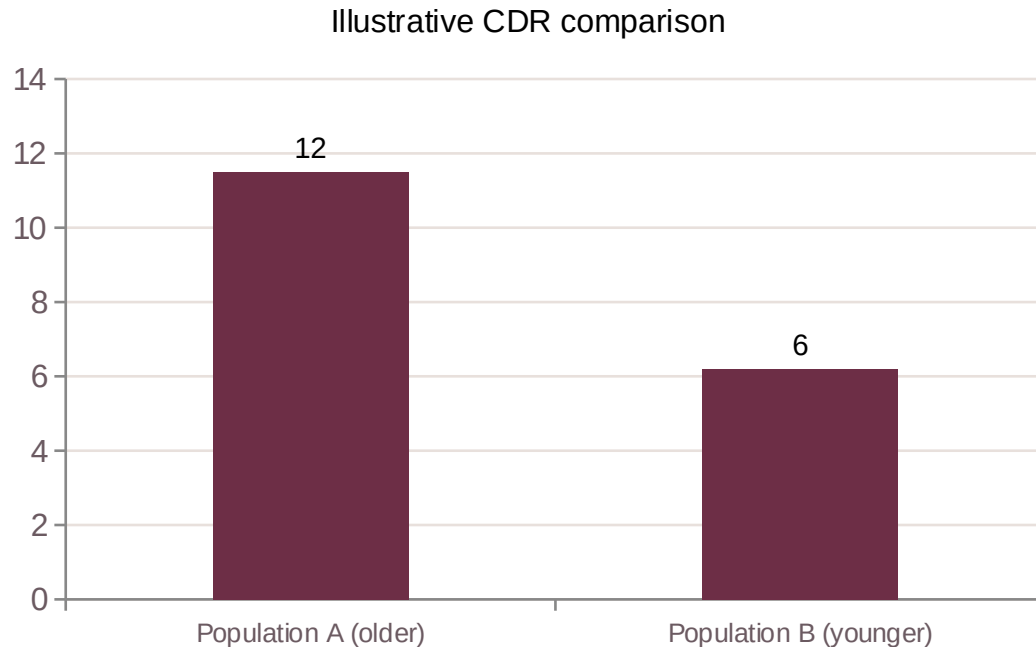
Crude Death Rate (CDR)

$$\text{CDR} = \frac{\text{Total number of deaths during the year}}{\text{Mid-year population}} \times 1,000$$

CDR is expressed as the number of deaths per 1,000 population.

A CLASSIC PARADOX

How Age Structure Distorts the Crude Death Rate



Population A and B are illustrative, for teaching purposes.

- Population A has an older age structure and a higher CDR (11.5). Population B is younger, with a lower CDR (6.2).
- Naive conclusion: "Population A is less healthy." This can be wrong — A's higher CDR may simply reflect having more people at high-mortality ages, not worse age-specific mortality.
- Solution: compare age-specific mortality rates directly, or standardize so age structure no longer drives the comparison (next slides).

Alternative measure

Age-Specific Mortality Rate (ASMR) & Age Standardization

Measure	Rates used	Population weights used	Main result
Age-Specific Mortality Rate (ASMR)	Observed rate for each age group	None	Mortality rate within age group (x)
Age Standardization			
- Direct standardization	Study population's ASMRs	Standard population	Age-standardized mortality rate
- Indirect standardization	Reference population's ASMRs	Study population	Expected deaths and SMR

The Age-Specific Mortality Rate (ASMR)

- Computed separately for each age group (usually 5-year bands) — removes the age-mixing problem within each band.
- The full set of ASMRs by age is the building block for life tables (see life-table session) and for both standardization methods on the following slides.
- Limitation: comparing dozens of age-specific rates across two populations, age group by age group, is informative but unwieldy for headline reporting — hence the need for a single summary number.

Age-Specific Mortality Rate (ASMR)

$$\text{ASMR}(x) = \frac{\text{Number of deaths among persons aged } x}{\text{Mid-year population aged } x} \times 1,000$$

Expressed as the number of deaths per 1,000 persons aged x .

HOW TO READ THIS CORRECTLY

Common Misconception: CDR vs. Age-Specific Mortality



MYTH

A lower crude death rate always means a healthier, lower-risk population.



REALITY

A lower CDR can simply mean a younger population. Two populations with identical age-specific mortality risk can post very different CDRs.



MYTH

If a country's CDR is rising over time, its health system must be getting worse.



REALITY

A rising CDR in an aging population (like Thailand's) can occur even while every age-specific rate is improving — the population is simply older on average.

The Age Standardization

1. Direct age standardization

Direct standardization applies the **study population's age-specific mortality rates** to the age distribution of a **standard population**. It produces an age-standardized mortality rate (ASDR). It is most appropriate when age-specific rates are reliable and when comparing several populations

$$ASDR = \frac{\sum(\text{Study ASMR}_x \times \text{Standard population}_x)}{\sum \text{Standard population}_x}$$

2. Indirect age standardization

Indirect standardization applies the **standard population's age-specific mortality rates** to the age distribution of the **study population**. It compares observed with expected deaths using the standardized mortality ratio (SMR). It is useful for small populations or when the study population's age-specific rates are unstable.

$$SMR = \frac{\text{Observed deaths}}{\text{Expected deaths}} \times 100$$

The **Standardized Mortality Ratio (SMR)** is a tool used to measure health inequality by comparing the actual number of deaths in a specific, vulnerable group or region to the number of deaths you would expect if that group had the same mortality rates as a standard reference population.

1. Direct Standardization: The Core Idea

Think of it as a "what-if" experiment

- 1 Take each population's own, real age-specific mortality rates (ASMRs) — these reflect true risk.
- 2 Ask: "What death rate would this population show if it had a common, shared age structure instead of its own?"
- 3 Apply each population's ASMRs to that one shared ("standard") age structure, and recompute the overall rate.
- 4 Because every population is now evaluated on the same age structure, any remaining difference reflects true mortality risk — not age mix.

Analogy: it's like comparing two classrooms' average test scores after adjusting so both classrooms have the same mix of weak and strong students.

1. Direct Age Standardization

- Requires: the age-specific mortality rates of the population(s) being compared, plus a standard population's age distribution (e.g., the WHO World Standard Population, or a national standard).
- Result: an age-standardized death rate (ASDR) — the death rate the population would have if it had the standard age structure.
- Best used when reliable age-specific rates are available for the population being studied — the common case for standardization in demography.

Direct Age-Standardized Death Rate (ASDR)

$$\text{ASDR} = \frac{\sum_x \left(\text{ASMR}_x^{\text{study}} \times \text{Standard population}_x \right)}{\sum_x \text{Standard population}_x}$$

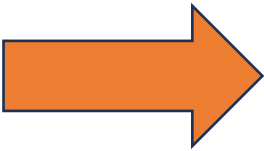
Direct Standardization: Two Populations

Step 1 — lay out each population's own ASMR by age, plus the shared standard weights.

Age group	Standard pop. (weights)	Pop. A ASMR /1,000	Pop. B ASMR /1,000
0-14	300	0.5	0.8
15-64	600	3.0	4.0
65+	100	45.0	60.0

- Notice: at every single age group, Population B's ASMR is higher than Population A's — B has genuinely worse mortality risk at every age.
- The standard population weights (300 / 600 / 100) are shared — they do not belong to either A or B; they exist purely to hold age structure constant.

Standard population weights and age-specific rates are illustrative, constructed for this worked example.



Direct Standardization: Calculating the ASDR

Age-Standardized Death Rate (ASDR)

$$ASDR_A = \frac{(0.5 \times 300) + (3.0 \times 600) + (45 \times 100)}{1,000} = \frac{6,450}{1,000} = 6.45 \text{ per 1,000}$$

$$ASDR_B = \frac{(0.8 \times 300) + (4.0 \times 600) + (60 \times 100)}{1,000} = \frac{8,640}{1,000} = 8.64 \text{ per 1,000}$$

Why this matters

- Even though Population B's age-specific rates are all higher than A's at every age, a naive crude-rate comparison (not shown here) could still favor B if B is much younger.
- After standardizing to the same age structure, B's ASDR (8.64) is clearly, fairly higher than A's (6.45) — a comparison no longer confounded by age.

WORKED EXAMPLE 2 • REAL DATA

Direct Standardization with Real Data: Thailand vs. Japan

Step 1 — *real, estimated age-specific mortality rates for Thailand and Japan (2024–2026), applied to the real WHO World Standard Population.*

Age group	WHO standard weight /1,000,000	Thailand ASMR /1,000	Japan ASMR /1,000
0–4	88,569	1.92	0.44
15–19	84,670	1.08	0.16
40–44	65,877	3.94	0.84
65–69	29,590	14.72	8.18
80–84	9,097	58.78	39.00
... (21 age groups total, 0–4 through 100+)	...	...	...

- At every age band shown, Thailand's estimated ASMR is well above Japan's — consistent with Thailand's lower life expectancy at birth (about 77.1 years vs. Japan's 85.2 years).
- Age-specific rates derived from single-year probabilities of death in each country's UN World Population Prospects life table (2026 revision).

Source: age-specific mortality derived from United Nations, World Population Prospects (2026 revision), compiled via GeoRank.org, <https://georank.org/life-expectancy/thailand> and [/japan](https://georank.org/life-expectancy/japan) (retrieved 2026-08-16).
Standard weights: Ahmad et al. (2001), WHO.

Thailand vs. Japan: Age-Standardized Death Rates

9.03

Thailand crude death rate
(per 1,000, 2024 — includes Thailand's own older age structure)

5.27

Thailand ASDR, standardized to
WHO World Standard Population (per 1,000)

2.62

Japan ASDR, standardized to the
same WHO standard (per 1,000)

Age-Standardized Death Rate (ASDR)

$$\text{ASDR} = \frac{\sum (\text{ASMR}_x \times w_x)}{\sum w_x}$$

where w_x is the standard population weight for age group x .

$$\text{ASDR}_{\text{Thailand}} = \frac{5,266,856}{1,000,000} = 5.27 \text{ per 1,000}$$

$$\text{ASDR}_{\text{Japan}} = \frac{2,616,400}{1,000,000} = 2.62 \text{ per 1,000}$$

- Once age structure is held constant (both evaluated on the same WHO standard), Thailand's mortality rate (5.27) is roughly double Japan's (2.62) — a genuine risk gap, not an artifact of Thailand simply having more older residents.
- Notice Thailand's own crude rate (9.03) is higher still than its standardized rate (5.27) — because Thailand's real population is older than the WHO world standard, inflating the crude figure beyond what age-specific risk alone would produce.

Direct or Indirect Standardization — Which One?

Use DIRECT standardization when...

- You have reliable age-specific rates for the population(s) you are studying.
- The population is large enough that age-specific rates are statistically stable (not based on tiny cell counts).
- You want a rate expressed in the same units as a crude rate (deaths per 1,000), directly comparable across many populations.

Use INDIRECT standardization (SMR) when...

- Age-specific rates for your study population are unreliable, unavailable, or based on small numbers (e.g., an occupational cohort, a small district).
- You know the study population's age structure and total observed deaths, and can borrow a standard population's age-specific rates.
- You are comparing one specific population against a reference standard, not several populations against each other.

2. Indirect Age Standardization

- Idea: apply a standard population's known age-specific mortality rates to your study population's own age structure, to get the number of deaths "expected" if the study population behaved like the standard.
- Used when the study population's own age-specific rates are unreliable or unavailable — e.g., a small population, or an occupational cohort — but its age structure and total deaths are known.
- Output: the Standardized Mortality Ratio (SMR) — observed deaths ÷ expected deaths, usually × 100. SMR > 100 means excess mortality relative to the standard.

Standardized Mortality Ratio (SMR)

$$\text{SMR} = \frac{\text{Observed deaths}}{\text{Expected deaths}} \times 100$$

where:

$$\text{Expected deaths} = \sum_x (\text{Standard ASMR at age } x \times \text{Study population at age } x)$$

An SMR of 100 indicates that observed mortality equals expected mortality.

WORKED EXAMPLE 1

Indirect Standardization: Computing an SMR

Age group	Study pop. (persons)	Standard ASMR /1,000	Expected deaths
0–14	4,000	0.6	2.4
15–64	9,000	3.5	31.5
65+	2,000	50.0	100.0
Total	15,000		133.9

Suppose this study population actually recorded 160 observed deaths in the year.

$$SMR = \left(\frac{160}{133.9} \right) \times 100 = 119.5$$

Interpretation

An SMR of 119.5 means this population experienced about 19.5% more deaths than expected if it had the standard population's age-specific mortality — evidence of excess mortality risk once age is accounted for.

In other words: An SMR of 119.5 means that the population had **19.5% more deaths than expected after accounting for age.**

Standard ASMRs, study population, and observed deaths are illustrative, constructed for this worked example.

SMR with Real Data: Thailand's Mortality vs. Japan's Standard

Question: if Thailand experienced Japan's age-specific mortality rates, how many deaths would be "expected" — and how does that compare with Thailand's real observed deaths?

Age group	Thailand population (thousands, 2024)	Japan ASMR /1,000 ("standard")	Expected deaths (thousands)
0–14	10,570.7	0.19	2.0
15–64	50,087.2	1.87 (avg.)	93.5
65+	11,010.2	23.0 (avg.)	253.2
21 five-year age bands (0–4...100+)	71,668.0 total	—	351.9 total

Broad-band rows above are illustrative summaries; the full calculation (and this workbook's practice exercises) use all 21 five-year age bands.

- Thailand's real, observed total deaths in 2024 were 647,349 (UN World Population Prospects 2024 revision).
- "Expected deaths" (351,896) is the count Thailand would show if every age group experienced Japan's much lower age-specific mortality, applied to Thailand's own population structure.

Sources: Thailand population — UN WPP 2024 revision via PopulationPyramid.net; Thailand observed deaths — UN WPP 2024 revision via Database.Earth, <https://database.earth/population/thailand/deaths>; Japan ASMR — UN WPP 2026 revision via GeoRank.org.

WORKED EXAMPLE 2 • RESULT

Thailand's SMR Relative to Japan's Mortality Standard

$$\text{SMR} = \frac{\text{Observed deaths}}{\text{Expected deaths}} \times 100$$

$$\text{SMR} = \frac{647,349}{351,896} \times 100 \approx 184$$

184

Thailand's SMR relative to a
"Japan mortality-risk" standard

+84%

Excess deaths relative to what
Japan's age-specific risk implies

77.1 / 85.2

Thailand / Japan life expectancy
at birth, years (2026 UN estimate)

Interpret

- An SMR of 184 means Thailand experienced roughly 84% more deaths in 2024 than it "should have" if every age group carried Japan's age-specific mortality risk — a genuinely large, real gap in underlying health risk, not an artifact of Thailand's different (younger) age structure.
- This is the indirect-standardization mirror image of the direct-standardization result on the previous slides (Thailand ASDR $\approx 2\times$ Japan's) — both methods point to the same substantive conclusion via different data requirements.

Common Misconceptions About Standardization



A standardized rate is the population's "real" death rate.

It is a hypothetical, constructed number — the rate the population would have under a specific standard age structure. The crude rate remains the real, observed rate; the standardized rate is a fair comparison tool.



Any two standardized rates can be compared, regardless of which standard was used.

Only rates standardized to the SAME standard population are comparable. An ASDR standardized to the WHO World Standard cannot be directly compared with one standardized to a national 2000 census standard.



An SMR of 100 always means "average" health.

It means average relative to whichever standard population's rates were used as the reference — a different standard (e.g., a healthier vs. sicker reference group) would shift what SMR = 100 represents.

EXAMPLE 1 • REAL DATA

Thailand's Mortality Snapshot, 2024–2026

9.0–9.1

crude death rate, per 1,000
population, 2023–2024

77.1

life expectancy at birth,
years, UN estimate 2026

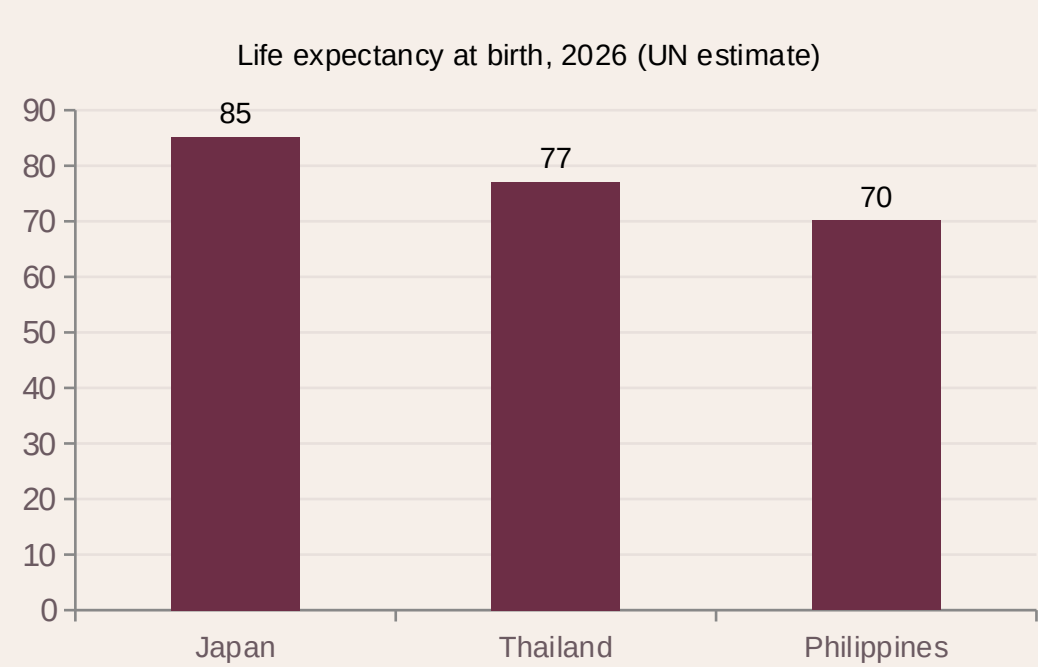
81.4 / 72.9

female / male life expectancy,
years (UN WPP 2026)

- Thailand's crude death rate looks moderate, but Thailand is also one of the most aged populations in the developing world (65+ share ≈15.4% of population, 2024 — Session 3).
- An older age structure mechanically pushes the CDR upward, even where age-specific survival is improving — exactly the distortion this block addresses methodologically, and exactly why the direct/indirect standardization worked examples above used real Thailand data.
- The large female–male life-expectancy gap (about 8.5 years) also motivates sex-specific and age-specific analysis rather than relying on a single aggregate figure.

Sources: Mahidol Population Gazette, Institute for Population and Social Research, Mahidol University (Vol. 33, Jan. 2024), CDR = 9.1; GeoRank.org (2026), compiling UN World Population Prospects 2026 revision, <https://georank.org/life-expectancy/thailand> (retrieved 2026-08-16).

Three Asian Countries, Three Transition Stages



Country	Life exp. (years)	Transition stage
Japan	85.2	Late/post-transition: aged, chronic-disease-dominant, very low fertility
Thailand	77.1	Late transition: rapidly aging, NCD-dominant, but still catching up to Japan
Philippines	70.2	Mid-transition: younger population, higher residual communicable-disease and maternal/child burden

- The 15-year life-expectancy gap between Japan and the Philippines illustrates why a single "Asia" mortality profile is misleading — countries occupy very different points on the epidemiological transition even within one region.
- Thailand sits between the two: its NCD burden already resembles Japan's, but its life expectancy has not yet closed the gap — a pattern the standardization worked examples above quantify directly.

Source: World Population Review (2026). Life expectancy in Asia 2026, compiling UN World Population Prospects and World Bank data. <https://worldpopulationreview.com/country-rankings/life-expectancy-in-asia> (retrieved 2026-08-16).

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IN-CLASS DISCUSSION

Choosing the Right Mortality Measure

Q1

A news report says "Country X has a lower death rate than Country Y, so it must have better healthcare."
What question would you ask before accepting this claim?

Q2

When would you prefer indirect standardization over direct standardization for a real analysis?
Use the Thailand/Japan examples to justify your answer.

Q3

Thailand's CDR is rising over time partly due to population aging rather than worsening health. How would you communicate this nuance to a non-technical policymaker?



BREAK

10 minutes · back at 1:10 for cause-of-death analysis and the COVID-19 retrospective

02

BLOCK 2

Cause-of-Death Analysis & the COVID-19 Mortality Shock

Moving from how many died to why — and what a global pandemic revealed about mortality measurement in real time.

Cause-of-Death Analysis

- Death registration records not just the fact and timing of death but an underlying cause, coded using the International Classification of Diseases (ICD-11) to enable comparison across countries and over time.
- Cause-specific mortality rate = $(\text{Deaths from cause C in a period} \div \text{Mid-year population}) \times 100,000$ — usually expressed per 100,000 to keep numbers readable for less-common causes.
- Proportional mortality = $(\text{Deaths from cause C} \div \text{Total deaths}) \times 100$ — useful for describing the mix of causes, but sensitive to the size of the denominator (total deaths), so it should not substitute for a population-based rate.



ICD-11

the WHO classification system coding
every cause of death worldwide

Years of Potential Life Lost (YPLL)

- YPLL weights premature deaths more heavily than deaths near the end of the natural lifespan — a death at age 20 contributes far more YPLL than one at age 74.
- This makes YPLL useful for ranking causes of death by their impact on years of life foregone, not simply by death counts — a cause with fewer total deaths but a younger typical age at death can rank higher in YPLL than a cause with more deaths concentrated at old age.
- Conceptually, YPLL is a precursor to YLL used in DALY calculations (Block 3) — both weight death by years of life foregone rather than treating every death identically.

Years of Potential Life Lost (YPLL)

$$YPLL = \sum_{i=1}^D (L - a_i)$$

where:

- L = selected reference age, such as 75
- a_i = age at death of person i
- D = number of deaths occurring before age L

Where Cause-of-Death Data Come From



Civil registration (CRVS)

Medically certified cause of death recorded at time of registration — the gold standard where coverage is complete and coding quality is high.



Verbal autopsy

Structured interview with family/caregivers after a death outside a health facility, used where medical certification is unavailable; algorithms assign a probable cause.



Modeled estimates

WHO Global Health Estimates and the Global Burden of Disease study combine registration, surveys, and statistical models to fill gaps — essential for global comparisons, but estimates, not direct counts.

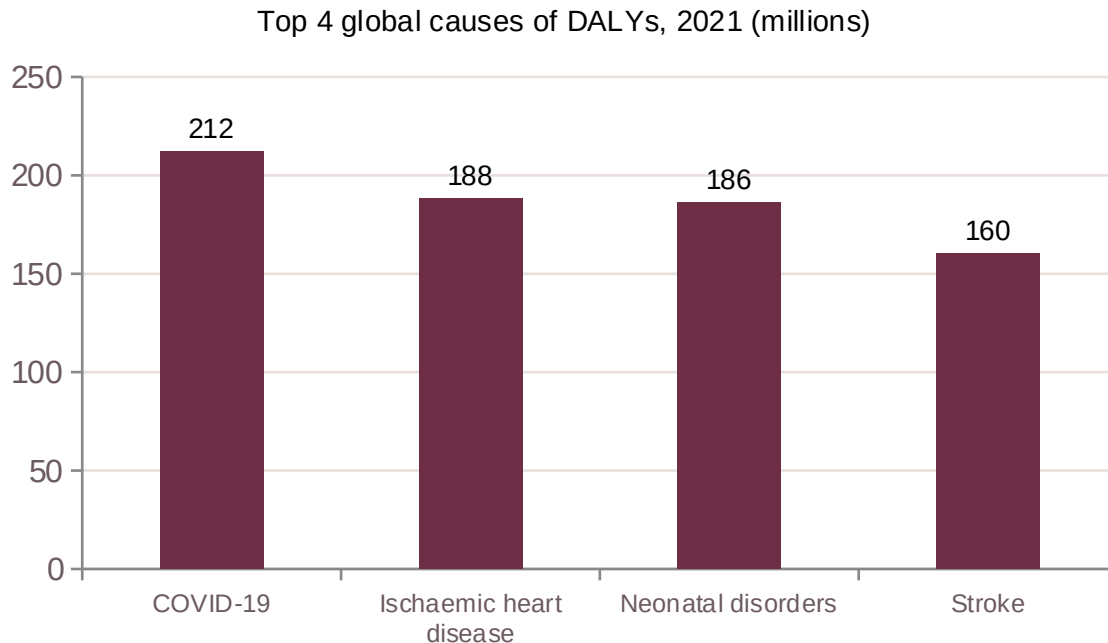
Implication: cross-country cause-of-death comparisons (such as GBD rankings, next slides) mix data of very different quality — always check a country's civil-registration completeness before over-interpreting small differences.

Data Quality: Garbage Codes & Misclassification

- **"Garbage codes"** (ill-defined or implausible causes, e.g., "old age," "cardiac arrest," or unspecified codes) are assigned when the true underlying cause could not be determined or was mis-recorded — WHO redistributes a share of these statistically before publishing cause rankings.
- **Verbal autopsy** — used where most deaths occur outside health facilities — depends on family recall and algorithmic classification; it performs well for some causes (e.g., maternal deaths, injuries) but poorly distinguishes among some chronic and infectious conditions with overlapping symptoms.
- **Coding-practice differences across countries** (which ICD version, how multiple-cause deaths are prioritized, coder training) can create apparent trends or country differences that reflect measurement change, not real epidemiological change.
- **Practical rule for consuming cause-of-death statistics:** always check whether a figure is a direct count from complete civil registration, or a modeled/adjusted estimate (e.g., GBD, WHO GHE) — and treat small year-to-year or country-to-country differences in modeled estimates with caution.

EXAMPLE 2 • REAL DATA (GBD 2021)

Leading Causes of Global Disease Burden, 2021



- 2021 is the first year in which COVID-19 became the single leading cause of global health loss — ahead of ischaemic heart disease, neonatal disorders, and stroke, the diseases that had dominated every prior year of the previous decade.
- This is a striking illustration of cause-of-death analysis in action: a single infectious disease reordered the global disease-burden ranking within about 18 months.

Source: GBD 2021 Diseases and Injuries Collaborators (2024). Global incidence, prevalence, YLDs, DALYs, and HALE for 371 diseases and injuries, 1990–2021: GBD 2021. *The Lancet*. [https://www.thelancet.com/journals/lancet/article/PIIS0140-6736\(24\)00757-8/fulltext](https://www.thelancet.com/journals/lancet/article/PIIS0140-6736(24)00757-8/fulltext)

The Post-Pandemic Picture: GBD 2023

2.80 bn

total global DALYs
estimated for 2023

~65%

share of global death and
disability from NCDs alone

8.4%

of all DALYs attributable to high
systolic blood pressure — the top risk
factor

- GBD 2023 (published 2025) estimated burden for 375 diseases and injuries and 88 risk factors across 204 countries — the pandemic-era peak of COVID-19 DALYs has receded, and the world's disease-burden ranking has reverted toward its pre-pandemic pattern: ischaemic heart disease, stroke, and diabetes now again dominate the top of the list.
- A key finding highlighted by IHME: overall mortality has continued to decline globally since 2021, but youth (adolescent and young-adult) deaths have risen in some regions and health inequities between the highest- and lowest-burden countries have widened rather than narrowed — a reminder that a falling global average can still mask worsening outcomes for specific groups.

Source: GBD 2023 Diseases and Injuries Collaborators (2025). Burden of 375 diseases and injuries, risk-attributable burden of 88 risk factors, and healthy life expectancy in 204 countries and territories, 1990–2023: GBD 2023. *The Lancet*. <https://www.sciencedirect.com/science/article/pii/S014067362501637X>; Institute for Health Metrics and Evaluation (2025). GBD 2023 findings summary. <https://www.healthdata.org/research-analysis/library/burden-375-diseases-and-injuries-risk-attributable-burden-88-risk-factors>

EXAMPLE 4 • REAL DATA, THAILAND

Thailand's NCD Burden, 2024–2025

76%

of all deaths in Thailand,
2024, attributable to NCDs
(~400,000 deaths)

75%

of Thai adults 60+ (≈7.4 million
people) live with a chronic NCD

USD 4.4 bn

estimated annual economic cost
of NCDs to Thailand (2025)

- Thailand's top NCD killers — diabetes, cardiovascular disease, cancer, chronic respiratory disease, and mental-health disorders — are driven by five behavioral risk factors: air pollution, smoking, unhealthy diets, physical inactivity, and alcohol consumption.
- Premature exit from the labor market accounts for 86% of Thailand's total NCD-related economic losses: 52% from premature deaths and 34% from early retirement due to illness — a cause-of-death pattern with direct workforce and productivity implications, not just health-system implications.
- This mirrors the epidemiological-transition pattern discussed earlier: Thailand's cause-of-death profile is already NCD-dominant like Japan's, even though its life expectancy has not yet closed the gap.

Source: Non-communicable diseases drive major economic losses in Thailand (2025, December 3). Khaosod English, reporting Thailand Ministry of Public Health / UN Thailand data.

<https://www.khaosodenglish.com/life/2025/12/03/non-communicable-diseases-drive-major-economic-losses-in-thailand/>

COVID-19 as a Population Health Shock

- COVID-19 gave demographers an unusually visible, globally synchronized mortality shock — with data published in near real time, contested in real time, and now revisited with several years of hindsight.
- Excess mortality — deaths above what historical trends would predict — became the preferred way to measure COVID-19's true toll, because it captures both direct COVID deaths and indirect effects (disrupted health services, delayed care, economic stress) without depending on inconsistent testing or cause-of-death coding across countries.
- By the mid-2020s, a substantial retrospective literature has moved past acute-phase reporting toward assessing lasting effects on life expectancy, health-system resilience, and whether mortality has fully returned to pre-pandemic trends.



2020–25

five years of evolving evidence,
from acute shock to retrospective
analysis

EXAMPLE 5 • REAL DATA

COVID-19 Erased a Decade of Life-Expectancy Gains

-1.8 yrs

global life expectancy at
birth, years, 2019→2021

-1.5 yrs

global healthy life expectancy
(HALE), years, 2019→2021

71.4 / 61.9

global life expectancy / HALE,
2021 (years)

- WHO's World Health Statistics 2024 report confirmed the pandemic reversed roughly a decade of global gains in life expectancy within just two years — global life expectancy fell to 71.4 years in 2021, back to its 2012 level.
- Healthy life expectancy (HALE) fell in step, to 61.9 years — also a return to 2012 levels.
- The decline was not evenly distributed: the Americas region saw HALE fall from 65.18 (2019) to 62.38 (2021), and by 2023 remained about 0.69 years below its pre-pandemic peak — recovery has been real but incomplete (see Example 8, Block 3).

Source: World Health Organization (2024). COVID-19 eliminated a decade of progress in global level of life expectancy. World Health Statistics 2024. <https://www.who.int/news/item/24-05-2024-covid-19-eliminated-a-decade-of-progress-in-global-level-of-life-expectancy>

Has Excess Mortality Really Ended?

- A 2025 ecological study in the International Journal of Epidemiology examined all-cause mortality across 21 countries and found sustained excess mortality persisting well beyond the acute pandemic phase — not a clean return to pre-2020 trends.
- Proposed explanations include: **lingering effects of COVID-19 infection (including post-COVID condition), delayed diagnosis and treatment of chronic disease during the pandemic, and disrupted health-system capacity that has not fully recovered in every country.**
- A separate 2025 US cohort study (JAMA Health Forum) similarly tracked excess deaths before, during, and after the pandemic, reinforcing that "post-pandemic" does not automatically mean "pre-pandemic baseline restored."
- Methodological lesson: excess-mortality estimates depend heavily on the choice of historical baseline and modeling method — a key critical-appraisal skill for consuming mortality statistics in the news.

Sources: Sustained excess all-cause mortality post COVID-19 in 21 countries: an ecological investigation (2025). *International Journal of Epidemiology*, 54(3).
<https://academic.oup.com/ije/article/54/3/dyaf075/8159616>; Excess US deaths before, during, and after the COVID-19 pandemic (2025). *JAMA Health Forum*, via PubMed.
<https://pubmed.ncbi.nlm.nih.gov/40408091/>



IN-CLASS DISCUSSION

Lessons from a Global Mortality Shock

Q1

Why might excess mortality be a more trustworthy cross-country measure of COVID-19's impact than official COVID-19 death counts?

Q2

What are at least two reasons excess mortality could remain elevated years after the acute pandemic phase ended?

Q3

How would you explain to a policymaker why 'life expectancy has recovered to 2019 levels' and 'excess mortality is still occurring' are not necessarily contradictory statements?

03

BLOCK 3 • 2:00–2:45

Summary Measures of Population Health

Combining mortality and morbidity into single indicators — DALYs, YLL, YLD, and healthy life expectancy — and the disability frameworks behind them.

WHY SUMMARIZE HEALTH

Summary Measures of Population Health

- Death rates alone miss a large share of the health burden: chronic illness, injury, and disability reduce wellbeing and productivity without necessarily causing death.
- Summary measures of population health combine mortality and non-fatal health loss into a single number, enabling comparison across diseases, countries, and time in one consistent unit.
- The methodological and ethical foundations for these measures — comparability, valuation of health states, and the policy uses and misuses of a single number — were formalized in a landmark WHO volume.



2002

the landmark WHO volume
defining today's summary health
measures

Source: Murray, C. J. L., Salomon, J. A., Mathers, C. D., & Lopez, A. D. (Eds.). (2002). *Summary measures of population health: Concepts, ethics, measurement and applications*. World Health Organization. <https://iris.who.int/handle/10665/42439>

THE CORE METRIC

Disability-Adjusted Life Years (DALYs)

- DALYs let a single unit combine dying young (YLL) and living with illness or impairment (YLD) — essential because these two burdens are otherwise not comparable.
- YLD requires a disability weight (0 = full health, 1 = death-equivalent) for each health state, derived from large population surveys asking people to compare health states — a deliberately value-laden, and openly debated, step (Murray et al., 2002).
- DALYs are the core metric of the Global Burden of Disease (GBD) study, used in Examples 2 and 3 of this session.

Disability-Adjusted Life Year (DALY)

$$\text{DALY} = \text{YLL} + \text{YLD}$$

where:

- YLL = years of life lost due to premature death
- YLD = years lived with disability or non-fatal health loss

One DALY represents one year of healthy life lost.

WORKED EXAMPLE 1

Computing a DALY Estimate — Road-Traffic Injury

Scenario

A person dies of a road-traffic injury at age 30. Standard life expectancy at age 30 is 50 more

$$\text{YLL} = 50 \text{ years}$$

Separately, 100 people in a population live with moderate hearing loss (disability weight = 0.02) for 1 year each

$$\text{YLD} = 100 \times 1 \times 0.02 = 2$$

years

$$\text{DALYs} = 50 + 2 = 52$$

Reading this result

Notice how a single premature death (50 YLL) can outweigh a much larger number of people living with a mild condition (2 YLD from 100 people) — DALYs make this trade-off explicit and comparable.

Real GBD calculations sum YLL and YLD across every age, sex, cause, and location — this is the same logic applied at national and global scale.

Computing a DALY Estimate — Type 2 Diabetes

Scenario

A person dies of complications of type 2 diabetes at age 58. Standard life expectancy at age 58 is 24 more years.

$$\text{YLL} = 24 \text{ years}$$

Separately, 40 people live with diabetes with complications (disability weight ≈ 0.15 , per GBD disability-weight sets) for an average of 8 years each.

$$\text{YLD} = 40 \times 8 \times 0.15 = 48 \text{ years}$$

$$\text{DALYs} = 24 + 48 = 72$$

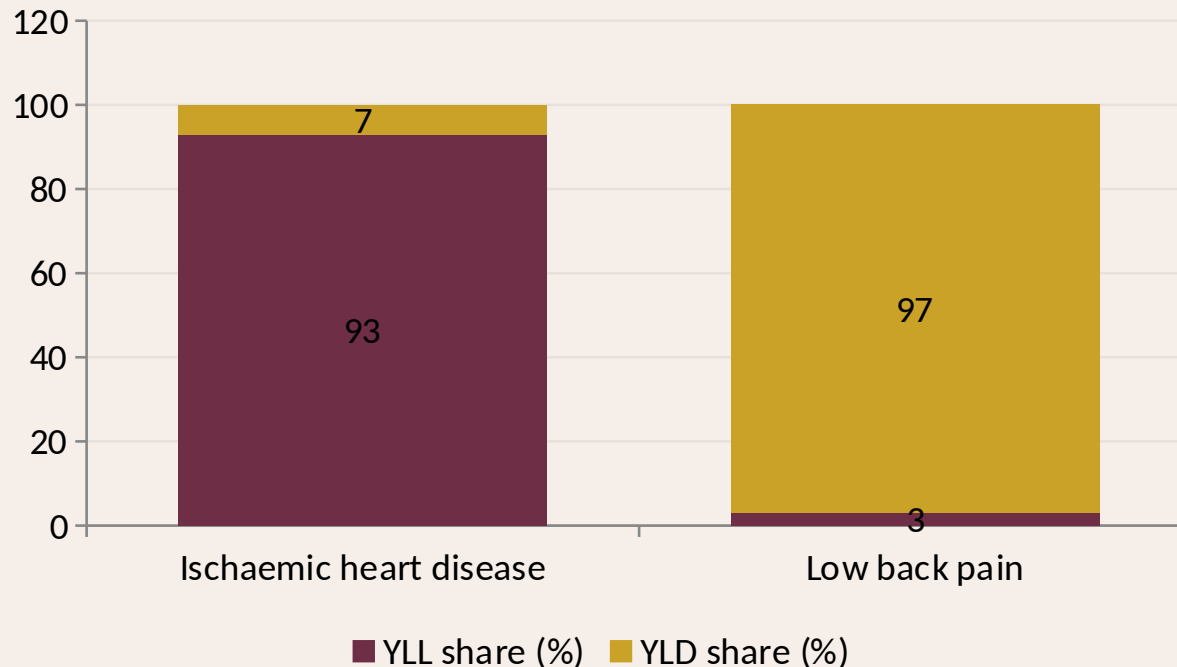
Reading this result

Unlike the road-injury example, here YLD (48) outweighs YLL (24) — this is characteristic of chronic NCDs like diabetes, which cause prolonged disability more often than they cause young, sudden deaths.

This is exactly why Thailand's rising NCD burden (Example 4) shows up strongly in DALYs even where mortality alone might look moderate: chronic disability accumulates over many years lived, not just at the moment of death.

YLL vs. YLD: Ischaemic Heart Disease vs. Low Back Pain

Approximate DALY composition by cause type



- Ischaemic heart disease is overwhelmingly a mortality-driven cause: in GBD accounting, roughly 90%+ of its DALYs come from YLL (premature death), because it kills relatively quickly once symptomatic and is a leading global cause of death (Example 2).
- Low back pain, by contrast, is almost entirely a disability-driven cause: it very rarely causes death, so essentially all of its DALYs are YLD — yet it remains one of the world's leading causes of disability, a fact invisible to any mortality-only statistic.
- This contrast is exactly why GBD reports DALYs (not just deaths): ranking causes by death counts alone would make low back pain disappear from the list entirely, even though it substantially reduces population wellbeing and productivity.

Figures are approximate, illustrative compositions consistent with published GBD YLL/YLD patterns for these two causes; exact shares vary by year, country, and GBD round.

Common Misconceptions About DALYs and HALE



A high-DALY cause is always a high-death-count cause.

Not necessarily — as the previous slide shows, a cause can rank high in DALYs primarily through YLD (disability), with very few deaths. Always check the YLL/YLD split before assuming "high DALYs" means "high mortality."



HALE and life expectancy always move in the same direction.

They usually move together, but not always by the same amount — a population can gain years of life expectancy while HALE grows more slowly (or even stalls), if those extra years are increasingly spent in poor health.



Disability weights are objective, universal facts about a disease.

They are survey-based value judgments about how much a health state reduces wellbeing relative to full health — reasonable people and cultures can and do disagree about them (Murray et al., 2002).

Functional Domains Behind Disability Measurement



Self-care

Bathing, dressing, eating, toileting without assistance.



Mobility

Walking, climbing stairs, moving from one place to another.



Communication

Speaking, understanding, seeing, and hearing well enough to interact.

- These functional domains — used in instruments such as the WHO Disability Assessment Schedule (WHODAS) and national disability surveys — describe what a person can and cannot do, independent of diagnosis.
- They feed directly into disability weights for YLD calculations: a health state that severely impairs mobility and self-care receives a higher disability weight than one that affects only, say, mild communication difficulty.

THE HEADLINE INDICATOR

Healthy Life Expectancy (HALE)

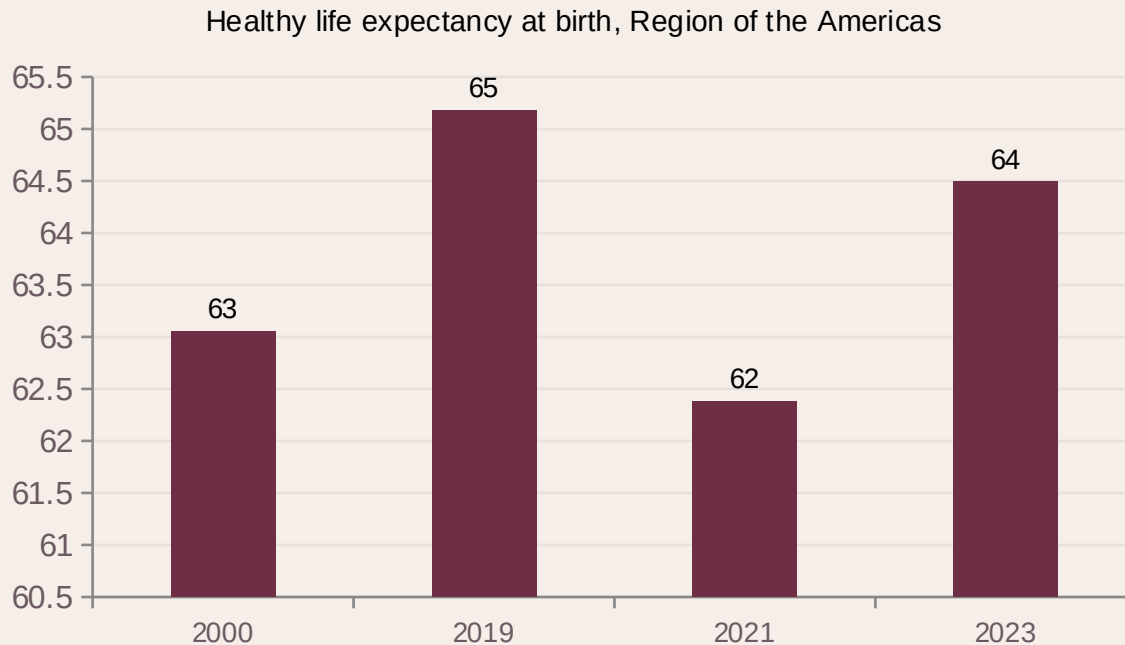
- HALE is conceptually life expectancy minus the equivalent years lost to disability — computed using the same disability weights that underlie YLD.
- It answers a different policy question than life expectancy alone: not just "how long do people live" but "how long do people live well."
- WHO and the GBD study both publish HALE by country, sex, and year, making it one of the most widely used single indicators of population health.

HALE

Average number of years a person can expect to live in "full health," accounting for time lived in less-than-full health due to disease or injury.

EXAMPLE 7 • REAL DATA

HALE Recovery Remains Incomplete in the Americas



- HALE in the Americas rose steadily from 63.05 years (2000) to a pre-pandemic peak of 65.18 years (2019), then fell sharply to 62.38 years during the acute pandemic phase (2021).
- By 2023, HALE had recovered to 64.49 years — better, but still about 0.69 years below the 2019 peak.
- This is the same "incomplete recovery" pattern seen globally in Example 5, now shown as a time series rather than a two-point comparison.

Source: Pan American Health Organization (2024/2025). Health-adjusted life expectancy (HALE). PAHO Enhanced Evidence and Intelligence for Action in Health. <https://pbdigital.paho.org/en/eob-2024-2025/impact-results/02-health-adjusted-life-expectancy-hale>

?

IN-CLASS DISCUSSION

Using Summary Measures Responsibly

Q1

Murray et al. (2002) stress that disability weights involve value judgments. What ethical concerns arise from assigning a single numeric weight to a health state?

Q2

Why might a country with rising life expectancy but flat HALE be a policy concern, even if headline "years of life" numbers look positive?

Q3

If you were advising Thailand's Ministry of Public Health, would you prioritize tracking DALYs, HALE, or cause-specific mortality rates for elderly-care policy? Justify your choice.

RECAP

Key Formulas at a Glance

Crude Death Rate

$(\text{Total deaths} \div \text{Mid-year population}) \times 1,000$

Age-Specific Mortality Rate

$(\text{Deaths at age } x \div \text{Population at age } x) \times 1,000$

Direct Age-Standardized Rate

$\Sigma (\text{Study ASMR} \times \text{Standard pop. weight}) \div \Sigma \text{Standard pop.}$

Standardized Mortality Ratio (indirect)

$(\text{Observed deaths} \div \text{Expected deaths}) \times 100$

YPLL

$\Sigma (\text{Reference age} - \text{Age at death}), \text{ summed across deaths below reference age}$

DALY

$\text{YLL} + \text{YLD}$

HALE

Life expectancy, adjusted for time lived in less-than-full health

Session 4 — Key Takeaways

- The crude death rate is easy to compute but confounds mortality risk with age structure; age-specific rates and standardization (direct or indirect) correct for this — as shown with both illustrative and real Thailand/Japan worked examples.
- Direct standardization needs the study population's own age-specific rates; indirect standardization (via the SMR) needs only its age structure and total deaths, applied to a standard population's rates. Both point to the same conclusion when applied to the same real data.
- Cause-of-death analysis, powered by ICD coding and studies like the Global Burden of Disease, shows that even a single pandemic year (COVID-19, 2021) can reorder the world's leading causes of health loss — and that recovery, per 2024–2025 retrospective evidence, has been real but incomplete.
- DALYs (= YLL + YLD) and HALE combine mortality and morbidity into single, comparable measures of population health, resting on the conceptual and ethical foundations laid out by Murray et al. (2002) — essential tools for any modern demographic analysis of health, including Thailand's rapidly growing NCD burden.



4

core skills from this session:
CDR/ASMR, standardization, cause-of-death, DALY/HALE

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04

CLOSING • 2:55–3:00 (ASSIGN AS HOMEWORK)

Practice Exercises

Eighteen questions spanning conceptual understanding and hands-on calculation — mortality-rate, standardization, cause-of-death, and DALY/HALE problems, several using this session's real Thailand and Japan data.

PRACTICE EXERCISES

Questions 1–5: Mortality Rates & Standardization Concepts

1. Define the crude death rate and explain why it can mislead when comparing two populations with different age structures.
2. What is the key difference between direct and indirect age standardization — and what data does each require?
3. A population of 250,000 records 2,150 deaths in a year. Calculate the crude death rate per 1,000.
4. Explain, in your own words, what a Standardized Mortality Ratio (SMR) of 85 would mean for a study population.
5. Why is proportional mortality (% of deaths from a given cause) not a substitute for a population-based cause-specific mortality rate?

Questions 6–10: Worked Calculations

6. Using this session's direct-standardization worked example (illustrative data), recompute Population A's ASDR if its 65+ ASMR rises from 45.0 to 55.0 per 1,000 (standard weights unchanged). Show your work.
7. A study population of 20,000 has 3,000 aged 0–14 (standard ASMR 0.5/1,000), 14,000 aged 15–64 (standard ASMR 3.0/1,000), and 3,000 aged 65+ (standard ASMR 48.0/1,000). Calculate expected deaths.
8. If the study population in Q7 actually recorded 210 observed deaths, calculate its SMR and interpret the result.
9. A person dies at age 45 from a cause with a standard remaining life expectancy of 35 years. Separately, 50 people live with a health state (disability weight 0.10) for 2 years each. Calculate total DALYs (YLL + YLD).
10. Explain how you would decide which age groups and how many (e.g., 5-year bands vs. three broad bands) to use when performing standardization for a real dataset.

PRACTICE EXERCISES

Questions 11–15: Cause of Death, COVID-19 & Summary Measures

11. Why did COVID-19 become the single leading global cause of DALYs in 2021, and what does the GBD 2023 update (Example 3) tell us about how the ranking has shifted since?
12. Distinguish 'excess mortality' from 'COVID-19 deaths' as reported in official statistics. Why do researchers often prefer the former for cross-country comparison?
13. Using the HALE time series from Example 7 (63.05 in 2000, 65.18 in 2019, 62.38 in 2021, 64.49 in 2023), calculate the percentage-point recovery from the pandemic trough (2021) to 2023, and the remaining gap to the 2019 peak.
14. Explain the conceptual difference between life expectancy and healthy life expectancy (HALE), and describe one policy scenario where the two measures would tell different stories.
15. Murray et al. (2002) argue that summary measures like DALYs involve ethical choices, not just technical calculation. Give one example of such a choice and discuss its implications.

Questions 16–18: Applying This Session's Real Data

16. Using the real Thailand and Japan direct-standardization worked example (this session), explain in your own words why Thailand's standardized rate ($\approx 5.27/1,000$) is lower than its own crude death rate ($\approx 9.0/1,000$), while Japan's standardized and crude rates are closer together.
17. Using the real SMR worked example (Thailand's observed vs. Japan-standard expected deaths), calculate what Thailand's SMR would be if its observed deaths had instead been 550,000 (all else unchanged). Interpret the change.
18. Using the Thailand NCD statistics (Example 4) and the YLL-vs-YLD decomposition (ischaemic heart disease vs. low back pain), argue whether Thailand's public-health priority should lean more toward reducing premature NCD deaths (YLL) or reducing years lived with NCD-related disability (YLD) — and what additional data you would want before deciding.

Thank you

Next session: continuing our study of demographic analysis

5140602 Fundamental Demographic Analysis · College of Population Studies, Chulalongkorn University · Asst. Prof. Dr. Ruttiya Bhula-or